

Advanced Microeconometrics - Machine Learning in economics

This part of the course provides an introduction to modern machine learning techniques, with a focus on applications in economics. It can be broken down in three parts. First, we introduce some of the most popular machine learning techniques and discuss their usage in *prediction* exercises (Part 1). Then, we discuss two recent applications of machine learning in (micro-)econom(etr)ics, where the goal lies with estimating *causal effects*. The first application concerns estimation of average treatment effects in the presence of *many* control variables (Part 2). The second application concerns estimation of heterogeneity in average treatment effects (Part 3).

The course is composed of lectures, which introduce the relevant “theory”, and corresponding lab sessions, where the “theory” is implemented in R. The evaluation is based on homework assignments and an in-class exam.

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Assignments

The material for the lab sessions and the assignments will be posted on the PSE konosys website. There will be three assignments, the first covering the material of sessions 1 and 2 (see below), the second covering the material of sessions 3 and 4, and the last covering the material of sessions 5–7. The assignments will be “practical”, i.e., they will require coding and consist in applying the methods you have learned to new datasets. You’ll have one week to work on each assignment. The exact dates on which they will be handed out is still to be determined. Assignments are to be handed in individually. Submission is via a link provided on the PSE konosys website.

In-class exam

The in-class exam will consist of true/false and multiple choice questions.

Detailed course outline and suggested reading

Digital copies of all references mentioned below are available on the course website. In what follows, the chapter references refer to James, Witten, Hastie, and Tibshirani (2013).

Session 1 – “Statistical learning” (Chapter 2.1)

– Model fit (Chapter 2.2.1)

– Cross-Validation (Chapter 5.1.1-4)

– Bias-Variance Trade-Off (Chapter 2.2.2)

See Hastie, Tibshirani, and Friedman (2009)—the “bible” on machine learning—for additional information on the above methods, including mathematical details. See also Varian (2014), which talks a bit more about “Big data.” An additional reference that “explains” machine learning to economists is Mullainathan and Spiess (2017).

Session 2 – Regularized linear regression (Chapter 6.2)

Mullainathan and Spiess (2017) touches upon the issue of “consistent model selection”, which is the topic of Part 2.

Sessions 3 & 4 considers the problem of estimating average treatment effects in the presence of *many* regressors. The first (and technical) reference is Belloni, Chernozhukov, and Hansen (2014b). An easily accessible explanation of their contribution is given in Belloni, Chernozhukov, and Hansen (2014a) (which also describes a closely related key contribution to the literature (Belloni, Chen, Chernozhukov, and Hansen, 2012)). We will focus on the double selection estimator of Belloni, Chernozhukov, and Hansen (2014b). Furthermore, we discuss the closely related double/debiased machine learning (estimator) proposed in Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey, and Robins (2018).

Session 5 – Trees and related machine learning algorithms (Chapter 8)

Session 6 considers the problem of estimating treatment effect heterogeneity. Technically, this amounts to (nonparametrically) estimating the conditional average treatment effect (CATE). We consider “causal trees” (Athey and Imbens, 2016) and “causal forests” (Wager and Athey, 2018). A simple exposition of their methods is given in Davis and Heller (2017).

Session 7 discusses an alternative approach that was proposed in Chernozhukov, Demirer, Duflo, and Fernández-Val (2025).

Session 8 In-class exam.

References

Athey, S. and G. Imbens (2016). Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences* 113(27), 7353–7360.

- Belloni, A., D. Chen, V. Chernozhukov, and C. Hansen (2012). Sparse models and methods for optimal instruments with an application to eminent domain. *Econometrica* 80(6), 2369–2429.
- Belloni, A., V. Chernozhukov, and C. Hansen (2014a). High-dimensional methods and inference on structural and treatment effects. *Journal of Economic Perspectives* 28(2), 29–50.
- Belloni, A., V. Chernozhukov, and C. Hansen (2014b). Inference on treatment effects after selection among high-dimensional controls. *The Review of Economic Studies* 81(2), 608–650.
- Chernozhukov, V., D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* 21(1), C1–C68.
- Chernozhukov, V., M. Demirer, E. Duflo, and I. Fernández-Val (2025). Fisher-Schultz lecture: Generic machine learning inference on heterogeneous treatment effects in randomized experiments, with an application to immunization in India. *Econometrics* 93(4), 1121–1164.
- Davis, J. and S. B. Heller (2017). Using Causal Forests to Predict Treatment Heterogeneity: An Application to Summer Jobs. *American Economic Review* 107(5), 546–50.
- Hastie, T., R. Tibshirani, and J. Friedman (2009). *The elements of statistical learning*. Springer.
- James, G., D. Witten, T. Hastie, and R. Tibshirani (2013). *An introduction to statistical learning*. Springer.
- Mullainathan, S. and J. Spiess (2017). Machine learning: an applied econometric approach. *Journal of Economic Perspectives* 31(2), 87–106.
- Varian, H. R. (2014). Big data: New tricks for econometrics. *Journal of Economic Perspectives* 28(2), 3–28.
- Wager, S. and S. Athey (2018). Estimation and Inference of Heterogeneous Treatment Effects using Random Forests. *Journal of the American Statistical Association* 113(523), 1228–1242.